# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A cyclist covers 120 m in 40 s along a straight road without turning back. The average speed is: 1

**(a)** 1.5 m $s^{-1}$ **(b)** 6 m $s^{-1}$ **(c)** 3 m $s^{-1}$ **(d)** 0.33 m $s^{-1}$

**2.** A speed of 36 km $h^{-1}$ in SI units is: 1

**(a)** 20 m $s^{-1}$ **(b)** 12 m $s^{-1}$ **(c)** 129.6 m $s^{-1}$ **(d)** 10 m $s^{-1}$

**3.** The velocity of a car changes from 10 m $s^{-1}$ to 30 m $s^{-1}$ in 4 s in a straight line. Its average acceleration is: 1

**(a)** 6 m $s^{-2}$ **(b)** 10 m $s^{-2}$ **(c)** −5 m $s^{-2}$ **(d)** 5 m $s^{-2}$

**4.** A body with initial velocity 5 m $s^{-1}$ moves with constant acceleration 2 m $s^{-2}$ for 8 s. Its final velocity is: 1

**(a)** 56 m $s^{-1}$ **(b)** 42 m $s^{-1}$ **(c)** 7 m $s^{-1}$ **(d)** 21 m $s^{-1}$

**5.** A body starts with velocity 0 m $s^{-1}$ and accelerates uniformly at 4 m $s^{-2}$ for 3 s. The distance travelled is: 1

**(a)** 9 m **(b)** 12 m **(c)** 18 m **(d)** 36 m

**6.** A child on a merry-go-round of radius 7 m completes one full revolution ($\pi = 22/7$). The distance travelled and the displacement are: 1

**(a)** 44 m and 14 m **(b)** 0 and 44 m **(c)** 44 m and 0 **(d)** 22 m and 7 m

**7.** The SI unit of acceleration is: 1

**(a)** m $s^{-2}$ **(b)** km $h^{-1}$ **(c)** m $s^{-1}$ **(d)** m $s^{2}$

**8.** The area under a velocity–time graph gives: 1

**(a)** displacement **(b)** acceleration **(c)** force **(d)** velocity

**9.** In uniform circular motion, the object has: 1

**(a)** constant velocity **(b)** zero acceleration
**(c)** changing speed **(d)** constant speed but changing velocity

**10.** A negative sign for acceleration in straight-line motion means: 1

**(a)** acceleration is opposite to the chosen positive direction
**(b)** the body is at rest
**(c)** the distance is negative
**(d)** the speed is zero

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** Distinguish between distance and displacement. 2

**12.** Can the magnitude of displacement be greater than the distance travelled? Give reason. 2

**13.** Define average velocity. State its SI unit. 2

**14.** A boy walks 3 m east and then 4 m north. Find the distance and the magnitude of displacement. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** Explain with an example why distance travelled and displacement can be different. 3

**16.** A student goes to the 4th floor of a building (3 m per floor) and then comes down to the 2nd floor. Find the vertical distance and displacement. 3

**17.** A man drives 200 km north in 3 h and then 200 km south in 2 h. Find the average speed and average velocity. 3

OR

Explain how a body can have zero average velocity but non-zero average speed. 3

**18.** A bus speeds up from 36 km $h^{-1}$ to 54 km $h^{-1}$ in 10 s. Find the acceleration. 3

OR

A bus moving at 54 km $h^{-1}$ is stopped in 5 s by braking. Find the acceleration and the stopping distance. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** (i) Define displacement and average velocity.
(ii) An athlete runs from O to A (100 m), turns back to B (60 m from A). Find the distance and displacement.
(iii) Compare average speed and average velocity. 5

OR

(i) Define acceleration with SI unit.
(ii) A car moves from 36 km $h^{-1}$ to 72 km $h^{-1}$ in 5 s. Find the acceleration.
(iii) Explain what the sign of acceleration means. 5

**20.** **Case study:** A bus at 36 km $h^{-1}$ accelerates for 10 s to 54 km $h^{-1}$, moves uniformly for a while, then brakes to rest in 5 s.
(i) Convert the speeds.
(ii) Find acceleration during acceleration.
(iii) Find acceleration while braking.
(iv) Find the braking distance. 5

OR

(i) How do you plot a position–time graph?
(ii) What does its slope represent?
(iii) Compare graphs for uniform and non-uniform motion. 5

## END OF QUESTION PAPER — Total: 40 Marks